

PaveLedger platform engineering design



Production architecture and implementation specification

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Design a road safety platform that converts camera observations into traceable investigations, routes the work through authorized teams and contractors, and verifies the outcome. The system must preserve uncertainty, original evidence and the complete history of each decision.

Design decision	Recommended approach
Capture	Managed Android capture app first; supported edge camera option; responsive web for office and contractors
Core service	Modular API service with background workers and a versioned workflow engine
Database	PostgreSQL with PostGIS for production spatial and tenant data
Media	Private regional object storage, redacted derivatives and time-limited authorized retrieval
Identity	Municipal OIDC or SAML SSO; MFA; server-enforced role and resource scope
AI	Purpose-trained road defect detector; human review and condition-specific evaluation
Hosting	Canadian primary region and Canadian recovery region when required by the customer

The delivered demonstration uses a private hosted web application and a persistent database with a compact state document. It exercises the workflow, not a production camera pipeline. It is a single-organization prototype, not the production multi-tenant architecture described here. Real AI inference, genuine evidence uploads, geospatial contract matching, contractor delivery, app-store capture apps and recovery infrastructure remain to be implemented.

Acceptance depends on field evidence and security tests, not the visual completeness of the demonstration.

System architecture and data flow

Layer	Components	Boundary
Vehicle	Camera, GNSS, clock, local detector, blur, encrypted queue	Device certificate and signed observation envelope
Ingestion	Upload authorization, quota guard, checksum validation, schema version	Authenticated tenant and device; no direct database exposure
Processing	Quality gate, inference, geospatial match, duplicate clustering	Asynchronous workers with bounded retry and dead-letter queue
Investigation	Ticket service, assignment, state engine, SLA and notification outbox	Server authorization on every read and mutation
Contract intelligence	Work scopes, warranty versions, contact directory, rule evaluation	Deterministic eligibility rules plus analyst confirmation
Evidence	Private objects, redacted copies, hashes, provenance, legal hold	Object retrieval checks the same tenant and ticket permissions
Experience	Admin web, analyst desk, contractor portal, mobile capture status	No data authorization delegated to browser controls
Operations	Metrics, audit archive, backups, restore and model registry	Separate production, test and development identities

Event order

A device uploads metadata and authorized evidence. The ingest service acknowledges only after durable storage is confirmed. Processing creates an observation and links it to a defect cluster. The case service opens one investigation or adds evidence to the existing one. Assignment creates an outbox event; a notification worker delivers the permitted alert.

After the analyst confirms the issue, the warranty engine proposes responsibility and the notice service applies the customer's dispatch policy. The contractor acknowledges, records the repair and supplies completion evidence. A later qualified pass or field inspection creates a verification observation. The reviewer closes or reopens the same case, preserving earlier repair attempts.

Integration strategy

Keep a stable internal road-segment ID and a mapping to the authority's GIS and work-order IDs. Publish a transactional outbox for integrations rather than updating an external system inside the ticket transaction. Reconcile acknowledgements and errors. The municipal asset system can remain the authoritative work-order system when the buyer requires it.

Vehicle capture and mobile applications

Use three product surfaces: a managed capture app for approved vehicles, a responsive analyst/admin web app, and a mobile-friendly contractor portal. A volunteer participant sees device health and participation controls, not other people's observations or municipal investigations.

Requirement	Engineering rule
Mount and power	Approved fixed mounting and power; do not obstruct vision or vehicle safety equipment
Image capture	Road-facing region of interest; configurable frame cadence; no audio or cabin recording
Location	GNSS fix, reported accuracy, heading, speed and capture timestamp stored per observation
Clock synchronization	UTC clock offset tracked; retain both capture time and receive time
Offline use	Encrypted bounded queue, resumable upload, retry jitter, checksum and acknowledgement
Privacy	On-device face and plate redaction where viable; server quality check before routine visibility
Device trust	Unique identity, signed configuration, key rotation, revocation and firmware version
Driver interaction	No manual interaction while moving; issue reporting only when safely parked
Volunteer controls	Opt-in zones and times; stop capture; no home or private-route tracking by default

Pilot a reliable Android phone or a dedicated camera/edge computer that exposes required timestamps and sensor data. A consumer dashcam with proprietary storage may be unsuitable even if it records good video. Validate thermal limits, vibration, glare, battery drain, mobile OS background restrictions and winter power behavior.

A dedicated stereo or depth sensor is a later option if measured pothole dimensions are required. A single ordinary camera image cannot reliably establish engineering depth or statutory severity dimensions without calibration and a validated method.

Keep full-route video off the standard upload path. Retain a short encrypted device buffer only if the authority authorizes it, and overwrite it according to policy. Store selected redacted evidence bundles centrally. Field onboarding must include fleet, labour and privacy review where applicable.

Computer vision and model governance

Detection pipeline

- Validate file type, hash, timestamp, sensor metadata and capture quality. Quarantine corrupted or malformed input.
- Reject or flag frames with excessive blur, occlusion, glare, snow cover or uncertain location. An unobservable road is not a defect-free road.
- Run a detector for potholes and selected crack classes. Keep class scores, bounding boxes or masks, model version and image dimensions.
- Group temporally adjacent frames into one observation bundle and estimate a candidate road location. Do not open a new ticket for every frame.
- Apply a separately approved prioritization rule using visible size cues, road class and exposure. Record measured dimensions only when actually measured.
- Route uncertain detections to review. Sample rejected frames and apparently clear road sections to estimate missed defects.

Training and evaluation

Public road-damage datasets can support an initial benchmark. The CRDDC dataset description includes longitudinal, transverse and alligator cracking and potholes. Dataset access does not itself establish commercial license rights or suitability for Canada; check dataset, code and model licenses independently. [S11]

Build local labelled data covering asphalt repairs, utility cuts, manholes, shadows, puddles, snow, wet surfaces, night lighting and different vehicles. Split training and test data by road segment, date and fleet so near-identical frames do not leak across sets.

Report event-level precision and recall, false alerts per driven kilometre, duplicate-ticket rate and location error. Show results by condition and damage class, with sample sizes and confidence intervals. A high frame-level accuracy on mostly undamaged roads can hide poor defect recall.

Use shadow deployment, a model registry, approval sign-off, rollback and drift monitoring. Store the version that produced each decision. A general language model may summarize analyst notes or extract candidate warranty clauses with citations, but cannot independently decide legal responsibility or close a safety investigation.

Geospatial accuracy and duplicate handling

Store coordinates in WGS84 with explicitly named latitude and longitude. Spatial calculations must use appropriate geography or projected coordinates, not treat degrees as metres. PostGIS supports distance queries in metres for geography through ST_DWithin. [S9]

Stage	Required method
Device position	Record GNSS fix and uncertainty at the image exposure time
Defect position	Use camera calibration and ground-plane or depth estimate where validated; retain method and uncertainty
Map match	Compare road geometry, direction, route, speed and heading; preserve alternatives
Lane assignment	Confirm from lane-level data or analyst review; parallel lanes are not interchangeable
Jurisdiction	Resolve authoritative road owner and maintenance responsibility, including boundary roads
Duplicate search	Search compatible nearby observations by spatial uncertainty, time, class and appearance
Cluster decision	Maintain observation IDs and one defect-cluster ID; allow reviewed split or merge

Do not label a vehicle GPS point as the exact pothole position. The pothole may be several metres ahead of the camera. Store the raw device position, estimated defect position, uncertainty radius and the calculation used. If the position is poor, ask for a field check before notifying a contractor.

Use a calibrated proximity threshold, initially 5 to 15 metres where the pilot supports it, combined with heading and same-lane evidence. This is an engineering starting range, not a universal rule. Nearby distinct potholes must remain separate. Do not merge opposite carriageways or stacked roads merely because coordinates are close.

A new sighting of an open defect adds an observation and may raise severity. A sighting near a closed case becomes a recurrence candidate. Confirm whether it is a failed repair at the same location or a different defect, then reopen the original or create a linked new case.

GIS geometry and road IDs change over time. Version the road network and retain the version used for a contract match so the original decision can be reconstructed.

Warranty and contractor responsibility engine

A camera cannot determine who repaired a road. Responsibility comes from the authority's contracts, work orders, permits, completion certificates, as-built extents and approved contact directory. Import and maintain those records as a separate controlled dataset.

Field group	Minimum data
Contract identity	Authority, contract ID, version, contractor legal entity and authorized subcontractor
Physical scope	Road segment, lane/direction, chainage or polygon, work type and limits
Coverage	Covered defect classes, exclusions, acceptance date, warranty term and extension rules
Notice rules	Trigger date, notice deadline, permitted channel, recipient and required attachments
Evidence	Signed document, clause reference, acceptance certificate and change orders
Contacts	Named POC, verified email/portal identity, escalation contact and effective dates

Rule evaluation

Evaluate tenant and jurisdiction, spatial work scope, work type, defect class, relevant dates, exclusions and notice rules. Return eligible candidate, ineligible, ambiguous or missing data, with the clause and input values that produced the result. Never infer eligibility from a contractor name and road name alone.

A warranty term can run from acceptance rather than invoice date. A defect noticed before expiry may have a different notice deadline. Keep observation, municipal awareness, confirmation and dispatch timestamps separately. Date semantics must be approved by the contract owner.

Controlled automation

Default to automatic ticket creation and reviewed contractor notices. Permit automatic notice only under a versioned municipal policy with validated geometry, approved terms, one unambiguous candidate, sufficient evidence and a verified contact. Log every rule and dispatch result. Ambiguity, missing documents, disputes and expired terms enter an analyst queue.

Unknown or uncovered responsibility routes to the authority's own maintenance process. A disputed warranty does not pause immediate safety action. The platform requests review and repair under a contract; it does not conclusively determine liability.

Ticket lifecycle and stage controls

Stage	Colour	Entry and exit evidence
Detected	Slate #54667E	Observation accepted; unique ticket or cluster link created
Team assigned	Blue #365CBF	Jurisdiction and responsible team selected
Analyst assigned	Indigo #584BC2	Authorized analyst selected; notification queued
Investigating	Violet #8055BD	Analyst starts review and records next action
Confirmed	Amber #A8640B	Issue is real; location and urgency reviewed
Notice pending	Orange #A14805	Approved draft exists; external dispatch still pending
Notice delivered	Orange #A14805	Provider confirms delivery state; acknowledgement still open
Acknowledged	Cyan #086E99	Contractor or municipal responder accepts receipt
Repair in progress	Teal #006D83	Work scheduled or started; target and owner recorded
Awaiting verification	Sea green #267764	Completion claimed with evidence; repeat observation requested
Verified closed	Green #167342	Eligible evidence and authorized reviewer accept repair
Reopened	Red #C32643	Failure or recurrence confirmed; earlier history retained
On hold	Grey #68656F	Reason, owner and review date required
Rejected	Grey #57606B	False report reason; evidence retained per policy
Duplicate	Grey #57606B	Canonical ticket link; observations preserved

The prototype combines notice preparation into one Notice prepared stage because it does not send external messages. Production must separate pending, failed, delivered and acknowledged events. Colour is always accompanied by a text label and accessible icon or description.

Transitions are server commands, not a free-form editable status field. Reject illegal jumps, absent evidence and unauthorized actors. Assignment may change without rewinding a confirmed investigation. Every transition stores actor, effective role, time, prior state, next state, reason and evidence references.

A hold never silently stops the safety clock. Rejected and duplicate reports require a reason; a duplicate must point to an existing accessible canonical case. Correcting a record creates a new audit event rather than overwriting earlier evidence.

Repair verification and reopening

Verification protocol

- When a contractor marks work complete, save the claimed completion time, work performed, before/after photographs and responsible person. This is a completion claim, not municipal acceptance.
- Schedule a repeat observation of the same road segment, lane and physical defect footprint after the reported repair time. Use a bus or other suitable vehicle when its route provides the required view.
- Check camera health, position, field of view, road-surface visibility and weather. Snow, a vehicle blocking the view or a different lane produces inconclusive evidence.
- Require positive visibility of the repaired area, a matched position and a reviewer decision. For critical or disputed defects, require field inspection or another independent observation according to policy.
- If the repair fails or the defect remains, reopen the case. Link the new evidence to a new repair attempt while retaining the original ticket ID, history and warranty record.

Observation outcome	System action
Clear view and acceptable repair	Present verification package for authorized closure
Clear view and defect remains	Reopen or retain repair state with failed-verification reason
No detection but road not visible	Keep awaiting verification; schedule another pass
Route never revisits	Escalate verification task to field inspector
New nearby defect	Create linked case after confirming it is distinct
Contractor photo only	Keep completion claimed until municipal verification rule is met

The next bus passage is a useful verification opportunity, not an unconditional closure trigger. A low-confidence absence is not evidence of repair. Repeated AI non-detection may supplement, but must not replace, a clear view and the approved acceptance rule.

Record reopen events separately from current status. Analytics should show currently reopened cases, cases ever reopened, number of repair attempts and reopened-after-closure rate using a clearly defined closure cohort. Avoid dividing lifetime reopens by this month's closures.

Role and resource authorization

Role	Allowed scope	Core capabilities
Platform operator	Service metadata; no routine tenant evidence	Provision tenant; support via logged, approved temporary access
Organization admin	Own authority	Manage memberships, teams, rules, integrations and all organization cases
Team lead	Assigned teams and geography	Triage, assign/reassign, supervise stages, team analytics
Analyst	Assigned cases or explicitly permitted shared queues	Investigate, propose responsibility, prepare notices and review evidence
Contract manager	Approved contracts and related cases	Maintain terms, approve ambiguous coverage and contractor contacts
Contractor POC	Cases assigned to their firm	Acknowledge, comment, provide repair updates and completion evidence
Auditor or executive	Approved organization scope	Read approved reports; export when authorized; no workflow mutation
Fleet or volunteer	Own device and approved capture context	Operate capture, inspect health, pause participation; no case browsing

Enforcement

Authorize the action against tenant, membership, role, team, geographic scope, assignment, contractor organization and sensitivity. Deny by default. Apply the same predicate to list endpoints, item endpoints, analytics, searches, exports, evidence downloads and background notifications. A hidden button is not an access control.

Production uses SSO identities and server-bound memberships, not a role supplied by the browser. Require MFA for privileged accounts, prompt offboarding and periodic access review. Analysts cannot grant themselves wider access. Contractors cannot close municipal cases, accept liability for another firm or browse unrelated contracts.

Support access is time-limited and approved. Administrative actions and impersonation, if offered at all, must be separately logged with the real and effective actor. The delivered owner-only role preview is a demonstration convenience and should be removed from a production municipal tenant.

Separation of duties

The contractor claims completion; an authorized municipal reviewer verifies it. High-risk warranty decisions may require a second approver. A platform operator should not be able to delete evidence and its audit history through ordinary support tools. Emergency access is exceptional, logged and reviewed.

Production data model

Entity	Important relationships and fields
Tenant	Authority, jurisdiction, region policy, retention profile
Identity and membership	SSO subject, tenant, role, effective dates, active flag
Team and team membership	Geography, lead, membership validity
Vehicle and device	Fleet owner, device key, calibration, firmware, health
Road network and segment	Versioned geometry, jurisdiction, road class, lane/direction
Observation	Device ID, event ID, capture/receive time, GPS, uncertainty, model version
Evidence object	Observation/ticket ID, hash, redaction version, storage key, legal hold
Defect cluster	Canonical point or geometry, class, uncertainty, related observations
Ticket	UUID, human-readable ID, cluster, state, severity, team, analyst, version
Assignment and status event	Append-only actor, timestamp, from/to, reason, correlation ID
Contract and scope	Version, contractor, covered geometry, work type, source document
Warranty rule and evaluation	Clause version, start/expiry/notice logic, candidate result and explanation
Contractor and contact	Legal entity, authorized POC, verified channel, effective dates
Repair attempt	Ticket, contractor, work start, completion claim and evidence
Verification	Matched observation, quality result, reviewer and decision
Notification and outbox	Recipient scope, idempotency key, retries, delivery and acknowledgement
SLA clock	Policy version, start event, due time, holds, escalation history
Audit event	Immutable event ID, real actor, action, resource, timestamp, request ID

Every tenant-owned key and index includes `tenant_id` where needed. Cross-tenant foreign keys must be impossible by schema design or verified server constraints. Prefer database row-level policies in addition to application authorization. Use UUIDs internally and an immutable human-readable ticket reference such as `PL-CAM-2027-000123`.

Store capture time and receive time separately, and retain original media metadata. Keep money in integer minor units with a currency code; location precision and severity confidence are different values. Index active team queues, assigned analyst queues, tenant/status/time, contract scope geometry and observation spatial/time searches.

The compact prototype state document is suitable for a bounded demonstration. Production must migrate to normalized entities, granular transactions, retention enforcement and a separately protected audit store before substantial field volume.

API design and event processing

Endpoint or event	Contract
POST /v1/observations	Device authentication; event UUID; metadata and object references; return durable receipt
GET /v1/tickets	Tenant-scoped filters and cursor pagination; no unrestricted aggregate count
POST /v1/tickets/{id}/assign	Team and analyst validated; expected_version; reason
POST /v1/tickets/{id}/transitions	Transition command with evidence IDs, expected_version and idempotency key
POST /v1/tickets/{id}/notices	Approved template, contract version and contact; enqueue once
POST /v1/repairs/{id}/complete	Authorized firm submits completion claim and evidence
POST /v1/tickets/{id}/verify	Matched repeat evidence, quality assessment and municipal reviewer decision
GET /v1/analytics	Same authorization scope as ticket lists; defined cohort and time grain

Example observation envelope

```
{ "schema_version": "1.0", "event_id": "device-generated-uuid", "device_id": "fleet-device-id", "captured_at":  
"2027-02-01T14:03:22Z", "latitude": 43.3605, "longitude": -80.3140, "horizontal_accuracy_m": 4.2,  
"model_version": "pothole-pilot-1", "evidence_sha256": "64-hex-character-digest" }
```

The server derives tenant from the device credential. It must not trust a client-provided tenant or contractor ID. Validate timestamps, numeric ranges, bounding boxes, object ownership, maximum body size and allowed content types before accepting a record.

Consistency and retries

Use at-least-once transport with a unique tenant/device/event key and idempotent consumers. In one database transaction, update the ticket version and append its status event and outbox record. Workers retry transient failures with exponential backoff and send permanent failures to a visible exception queue.

Use expected_version or ETag to prevent lost updates. Return 409 on conflicting edits and preserve the analyst's unsaved note. Return 401 for absent identity, 403 or non-enumerating 404 for forbidden resources, 422 for invalid evidence and 429 with retry guidance for quota pressure.

Do not let a slow email provider block a ticket transaction. Preserve message IDs, template versions, redacted attachment hashes, send attempts, delivery outcomes and acknowledgements. Delivered email is not proof of contractor acceptance.

Contractor notices and escalation

Notice contents

- Unique ticket ID, road authority, defect class, risk priority and case owner.
- Redacted issue image, capture date and time, coordinates, accuracy radius, road segment, lane/direction and practical access description.
- Repair contract reference, work-scope match, applicable warranty clause and notice basis.
- Requested acknowledgement and response dates derived from approved policy or contract terms.
- Secure scoped portal link, contact for disputes and clear statement of whether urgent municipal action has already occurred.

Dispatch process

The service resolves the POC from an approved contact directory; it does not guess or scrape an email address. Preview the notice and record the approver unless an approved automatic rule applies. Send from the authority or an explicitly delegated service identity, with accessible text and a secure evidence link.

Use a transactional outbox and idempotent message key so retries cannot send duplicate demands. Failed delivery goes to an operations queue. Acknowledgement, responsibility accepted, responsibility disputed and repair scheduled are separate events. Do not interpret an email open as acceptance.

Escalations

A safety escalation goes to the municipal duty process when the hazard is urgent or the local deadline requires action. A contractor escalation follows the contract notice path. A verification escalation occurs when no suitable vehicle revisits. These clocks run independently.

Configure working calendars, time zones, after-hours coverage and service levels per authority. Store deadlines in UTC while showing the user's local time. Record policy versions and clock-start events. Holds need reasons and review dates; legal or safety clocks pause only if the governing rule permits it.

Demonstration limitation

The delivered website can prepare a downloadable draft notice and creates in-app assignment notifications. It does not send email, SMS, push notifications or requests to real contractors. The example firm names and .invalid addresses are fictional. No external recipient has been contacted.

Privacy security and evidence integrity

Complete a privacy impact assessment and legal-authority review before real camera collection. For Ontario buyers, use current applicable public-sector privacy law and commissioner guidance. The guidance page itself flags recent and upcoming changes; do not rely on a static 2015 document alone. [S6]

Control	Design requirement
Collection limits	Road-facing purpose; no audio, cabin recording, face recognition or unnecessary plate history
Redaction	Blur faces and plates before ordinary access; retain restricted originals only when lawful and necessary
Retention	Customer-approved schedules by data class; legal hold overrides deletion; deletion logs
Encryption	TLS in transit; managed keys at rest; least-privilege key access and rotation
Identity	SSO, MFA, short-lived tokens, session controls, offboarding and device revocation
Isolation	Tenant scope in database, queues, objects, caches, search and analytics
Evidence integrity	Cryptographic hash, original metadata, derivative lineage, append-only history and controlled corrections
Downloads	Authorization checked before signed URLs; short expiry; no public evidence bucket
Operations	Secrets manager, dependency scanning, tested backup, incident process and monitored admin access

Proposed starting retention profile for review: device raw buffer up to 24 hours, unredacted quarantine up to 72 hours if justified, ordinary redacted observation media 90 days, and ticket evidence according to the authority's approved records schedule. These are design proposals, not legal retention rules. A legal hold may retain selected evidence longer.

Evidence records should bind the image hash to the capture event, device identity and software version. A hash proves that bytes have not changed; it does not prove the original scene was genuine. Device attestation, independent repeat observations and analyst review address different provenance risks.

Canadian-region hosting should include the database, media, backups and processing. Also review global telemetry, content delivery, support tooling and external AI services. The hosted demonstration does not claim a certified Canadian-only data boundary.

Reliability scale and cost controls

Proposed target	Definition and qualification
Availability	99.9 percent monthly API target for production; excludes no implicit legal deadlines
Online ingest	95th percentile metadata acceptance under 2 seconds at agreed pilot load
Investigation availability	95th percentile under 60 seconds after a valid online bundle; offline delays tracked separately
Queue recovery	No silent event loss; retry and dead-letter processing visible to operators
Recovery point	Production target of 15 minutes or less, proven by recovery tests
Recovery time	Production target of 4 hours or less for core investigation access
Interactive list	95th percentile under 2 seconds for indexed authorized queues
Accessibility	Design and test toward WCAG 2.2 AA; buyer-specific legal requirements separately confirmed

Illustrative storage sizing

At 100 vehicles, 100 route-km per day and 20 uploaded bundles per route-km, the service receives 200,000 bundles daily. At 0.6 MB per bundle that is approximately 120 GB/day or 3.6 TB over 30 days, before versions and backups. These are workload assumptions; pilot observations must replace them. Raw video at 4 Mbit/s for eight hours is about 14.4 GB per vehicle/day, or 1.44 TB/day for 100 vehicles. Selective upload materially changes cost.

Use an edge quality gate and upload accepted evidence plus a governed negative sample. Keep sampling sufficient to assess false negatives. Do not discard data merely to make precision appear higher. Throttle bulk transfers during cellular congestion while retaining urgent metadata and queue status.

Capacity and resilience

Use stateless APIs, a durable queue, separately scalable inference workers and partitioned observation data. Begin with a modular service rather than many small services. Separate GPU inference when workload justifies it. Partition by tenant and time, with appropriate spatial indexes.

Track cost per route-km, bundle, reviewed case and paying authority. Load-test bursty bus depot uploads, storage limits and downstream outages. Targets are objectives for production validation; no service level is guaranteed by the prototype. AWS documents Canadian regions, but final service availability, residency and cost must be checked for the selected deployment. [S10]

Analytics and analyst experience

Investigation desk

The analyst opens an assigned bucket sorted by risk and due action. A case shows the original issue and redacted evidence, precise recorded coordinates with uncertainty, road/lane and jurisdiction, repeated sightings, AI version, contract candidates, notes, assignments, deadlines and the full timeline. Compare before and after evidence side by side. Preserve notes during errors or concurrent edits.

The next-action panel should expose only permitted transitions and explain missing evidence. Urgent safety tasks remain visible during warranty disputes. The mobile contractor view focuses on acknowledgement, repair schedule, completion evidence and messages.

Measure	Definition
Current backlog	Open canonical tickets by stage, age, severity and responsible team
Triage time	First human triage timestamp minus receipt; report median and 95th percentile
Time to repair claim	Completion claim minus confirmed issue time; distinguish dispatch and mobilization
Verification delay	Accepted verification time minus completion claim
Closure cohort reopen rate	Closed tickets later reopened within a defined window / eligible closure cohort
Warranty recovery	Accepted and evidenced avoided costs, separate from candidate claim value
Coverage freshness	Time since eligible visible observation for each managed lane segment
Detection quality	Event precision, sampled recall, false alerts per route-km and location error
Operational health	Device uptime, stale feeds, failed uploads, queue lag and redaction failures

All aggregates use the same tenant, team, assignment and contractor scopes as case access. An auditor may see organization-wide statistics only if explicitly authorized. Contractors see their own repair performance and assigned cases, not competitor data.

Show labels and icons with colour. Provide keyboard navigation, clear focus, readable contrast, screen-reader labels, accessible tables and an alternative to map-only controls. Do not make severity rely on red/green distinction.

Production mapping requires an authoritative basemap/license choice and a geospatial service. The delivered prototype shows coordinates in case detail and does not pretend that its sample locations are surveyed defect positions.

Implementation plan and acceptance tests

Work package	Planning effort	Acceptance evidence
Discovery and contract model	3 to 5 person-weeks	Requirements, sample data, authority mapping and privacy plan
Platform and permissions	8 to 12 person-weeks	Isolation tests, workflow engine, audit and assignment
Capture and ingestion	8 to 12 person-weeks	Offline recovery, signed events, field mount and clock tests
Vision and location	10 to 16 person-weeks	Local benchmark, negative sampling and surveyed location error
Warranty and notices	6 to 10 person-weeks	Clause review, ambiguous cases, idempotent dispatch
Verification and analytics	5 to 8 person-weeks	Clear-pass gate, reopen history and scoped cohort reports
Hardening and operations	8 to 12 person-weeks	Pen test, accessible flows, restoration and load test

Total planning effort is 48 to 75 person-weeks before material procurement delays or additional integrations. With parallel work by a small competent team, plan a first field pilot over four to six months and production hardening thereafter. These are engineering estimates, not fixed bids.

Tests that must pass

- Tenant A cannot list, fetch, search, export or download Tenant B records, including guessed IDs and signed-object paths.
- A contractor cannot close a case, read another firm's job or change responsibility. An analyst cannot create an admin membership.
- Two simultaneous assignments produce one accepted update and one visible conflict without losing notes. Replayed upload and notice events are idempotent.
- A nearby pothole in another lane remains distinct. Bad GPS produces uncertainty rather than false precision.
- An expired warranty, overlapping scopes, missing acceptance date and disputed responsibility never silently trigger a demand.
- No repair closes when the next pass is snowy, occluded, in the wrong lane or before the claimed repair time.
- Reopening preserves original ID, repair attempts and earlier evidence. Analytics distinguish current reopens from historical ones.
- Backup restoration recovers cases, media references, memberships and audit events to the agreed recovery point.

Prototype guide and production gap

Review the demonstration

- Open the workspace and review the attention queue and stage counts. Open King Street East to inspect the image, coordinates, evidence caveats and history.
- Use Assignment to choose a team and analyst. Save to create an in-app alert. Preview Avery Chen to see the analyst's assigned scope.
- Move an assigned case through Investigating and Confirmed, adding a reason at every change.
- For a King Street East case, select its active demonstration contract and confirm scope before preparing a notice. Download the draft to inspect its contents. No message is sent.
- Move a repair to Awaiting verification. The demonstration requires explicit reviewer confirmation before Verified closed. Reopen it to see the retained history and analytics change.
- Preview a contractor or auditor to inspect restricted actions. As admin, create a team or add a membership record. No invitation is sent.
- Reload the page to confirm saved records persist. Use search, stage filters and CSV export to review the permitted cases.

Implemented in this package	Requires production development
Private access and persistent single workspace	Customer SSO, multi-tenant onboarding and enterprise access lifecycle
Server-enforced role checks and scope filters	Fine-grained geographic policies and independent security validation
Unique IDs, assignments, stage history and in-app alerts	Durable normalized audit store, clocks, escalation and notification integrations
Contract examples and reviewed draft notice	Real contract ingestion, spatial matching, approval policies and external delivery
Illustrative evidence and verification action	Live inference, signed fleet capture, real media upload and matched repeat passes
Responsive web and scoped basic analytics	Managed mobile capture app, mapping, production metrics and billing

The prototype is intended for concept review with fictional data. It should not receive live municipal evidence or be presented as an operational road-inspection service. The production rollout gate is completion of the architecture, field validation and customer-specific controls described in this specification.

Technical and policy references are listed in the business plan as S5, S6 and S9-S11.